

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

ATTACHMENT B

DESIGN SPECIFICATIONS SECTION 723.480, 3.07

(4 Pages)

ADDENDUM NO. 12

7. Variation from Level: Plus or minus $\frac{1}{2}$ inch in any 40 foot run.

D. Performance Requirements:

1. Conduct inspections, perform testing, and make repairs or replace unsatisfactory precast units as required.
2. In addition to the above, in-place precast units may be rejected for any one of the following:
 - a. Exceeding specified installation tolerances.
 - b. Damaged during construction operations.
 - c. Exposed-to-view surfaces that develop surface finish deficiencies.
 - d. Other defects as listed in PCI publication MNL-117.

3.06 ADJUSTING AND CLEANING

A. Cleaning:

1. Perform samples of actual cleaning methods on precast mock-up in the presence of the Engineer.
2. Perform sample cleaning on mock-up after being in place a minimum of 30 days and with protection of minimum 50 percent of panel from cleaning processes to allow comparison between clean and dirty portions as directed by the Engineer.
3. Protect precast units before and after cleaning to prevent any spillage, rust stains, and other contaminants from penetrating or hardening on finished surfaces.
4. Immediately after completion of erection of precast units, clean exposed surfaces by brushing to remove loose particles, dirt, and dust.
5. Wash exposed surfaces with the cleaning method found satisfactory by the sample cleanings, and thoroughly rinse with clean water. Do not allow cleaning agents to enter interim and final drainage systems.

3.07 INSTALLATION OF ADHESIVE TYPE CEILING ANCHORAGE SYSTEM

- A. Comply with adhesive anchor system manufacturer's printed installation instructions and recommendations.
- B. Use a non destructive procedure acceptable to the Engineer to locate concrete reinforcement, embedded utilities, conduits, etc. in tunnel structure.
- C. Use only drilling systems and drill sizes specifically recommended by the anchor manufacturer.
- D. Core Drilling Through Embedded Reinforcing Bars.
 1. The Contractor shall be allowed to core drill into the tunnel roof slabs and through the embedded reinforcing bars in accordance with the following procedures:

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- a. Should rebar interference be encountered during the course of normal initial drilling operations, the hole in question shall be identified with paint. The Engineer shall be notified and shall verify if the interference is a rebar or another object such as but not limited to a conduit.
 - b. Abandon the first hole and drill a new hole 3 inches away from the initial location. If there is an interference on the second attempt notify the Engineer who again will verify the type of interference.
 - c. If the first hole contained a rebar the Contractor may proceed with core drilling through the initial drilled hole upon receiving authorization from the Engineer.
 - d. Should the interferences not be reinforcing bars the Engineer shall provide the Contractor with direction on how to proceed with the hole drilling.
 - e. Core drilling shall be perpendicular to the roof slab surface.
 - f. Maintain hole diameter as required by anchor manufacturer.
 - g. Core drill through entire section of the embedded rebar. Provide additional adhesive and threaded rod when depth of hole exceeds typical depth.
 - h. Document all core drilling operations in a Core Drilling Report which shall include at a minimum the locations of all core drilled holes, the depth and diameter of all core drilled holes, and the number, and size of all embedded rebar which were cored. Include a plan which documents the location of each cored hole.
 - i. Fasteners required to anchor drilling equipment to tunnel roof shall not cut reinforcing steel or other embedded items.
- E. Thoroughly clean bolt holes after drilling, with fresh, clean water or forced air as recommended by anchor system manufacturer.
- F. Patch all abandoned holes, including holes left from attaching the drilling equipment and anchor testing equipment to the tunnel roof, with epoxy grout compatible with epoxy coating of reinforcing steel.
- G. The following testing procedure is only for anchorage systems embedded in the concrete tunnel structure for the hanging ceiling supports. After chemical adhesive has cured, proof test all anchors to a service load of 3.25 kips tension per ASTM E 448, unconfined static tests for tension. Perform tension only test. Record results of proof tests and submit to Engineer. Clearly mark and abandon anchors which fail test. Proof test procedure must be submitted to and reviewed by Engineer prior to installation. The Contractor shall use an independent testing agency, approved by MHD.
- 1. Adhesive Anchor Rod Testing Procedure - 100% of Ceiling Anchors
 - 2. Repairs to concrete surfaces required due to Anchorage System testing shall be made, at no additional cost to the Department. Repair materials must be acceptable to the Engineer.
 - 3. Fasteners to anchor test equipment are prohibited.

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- H. Under no circumstances shall an anchor be installed within the embedment length of the face of the electric pull boxes or similar embedments.
- I. Engineer may conduct spot checks on anchors.
- J. Fully torque bolt to manufacturers specifications.

3.08 EXCESS GROUT

- A. Handle, transport, and dispose of excess grout in accordance with Massachusetts Department of Environmental Protection solid and hazardous waste regulations.

4.02 BASIS OF PAYMENT

(replace Subparagraph 4.02A with the following:)

- A. The Work of this Section will be paid for under Section 996.910 - Tunnel Finishes.

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